

Mohammad hadi Niknam

Email | LinkedIn | Personal Website

EDUCATION

– Amirkabir University of Technology (Tehran Polytechnic)

2023 – February 2027

Bachelor of Science in Computer Science

220 ECTS – 7th Semester – CGPA: 16.60/20

* **BSc Thesis:** *Human-Adaptive Safe Reinforcement Learning for Medical Ultrasound Robotic Arms*

RESEARCH INTERESTS

- **Reinforcement Learning** : RL Theory, LLM Alignment, Decision Making Under Uncertainty, MultiAgent RL
- **Medical Robotics:** Robot learning, human-adaptive control, ultrasound-guided robotics
- **Medical AI:** Medical imaging, foundation models, clinically robust AI

SELECTED PUBLICATIONS AND RESEARCH OUTPUTS

1. **SRIQL: Implicit Q Learning for Ultralight-weight Image Super Resolution.** Farzan Niknejad, Mohammad Hadi Niknam, and Javad Alirezaie. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026. Manuscript.
2. **URLSR: Efficient Image Super Resolution Network using Neural Operations and Reinforcement Learning.** Farzan Niknejad, Mohammad Hadi Niknam, and Javad Alirezaie. *BMC Medical Informatics and Decision Making*, 2026. Manuscript.
3. **Fairness-Aware Operating-Room Sequencing for Acute Orthopaedic Trauma: A Retrospective Capacity-Calibrated Stress Test Using MOVER.** Mohammad Hadi Niknam, Saeed Vosoughi, Ghazaleh Shakiba Maram, et al. *BMC Medical Informatics and Decision Making*, 2026. Manuscript.
4. **Human-Adaptive Safe Reinforcement Learning for Medical Ultrasound Robotic Arms.** Mohammad Hadi Niknam. BSc Thesis, Amirkabir University of Technology (Tehran Polytechnic), 2026.
5. **Arena Human Preference 90K: Engineered Features and Correctness Annotations.** Nazanin Mirzaei, Parsa Naderi, Mohammad Hadi Niknam, et al. Dataset, 2026. [Dataset]
6. **Imaging-Based Artificial Intelligence for Osteonecrosis of the Jaw** Mohammadreza Dolikhani; Mona Gorji; Mohammad Hadi Niknam et al. Manuscript under review Artificial Intelligence In Medicine, 2026.

RESEARCH EXPERIENCE

- * **Dr.Javad Alirezaie Lab** *Toronto Metropolitan University - Canada - Remote*
Research Intern July 2025 - September 2026
 - Super Resolution field, Designed RL-based NAS over lightweight blocks, evaluate it on DIV2K,Set14,Set5,Urban100,
- * *Increase PSNR 1db over*
Research Project Coordinator September 2025 - Now
 - Led a 3-member undergraduate research team on efficient image super-resolution, coordinating paper discussions, experimental milestones, task allocation, and research deliverables.

- * **Dr.Nassir Navab** *Computer Aided Medical Procedures | TUM | Munich - Germany*
Research Intern May 2025 - Now
 - Review Vision language models on Ultrasound modality, Investigating role of RL in ultrasound modality

- * **Shafa Yahyaian Hospital, Iran University of Medical Sciences** *Tehran, Iran*
Research Assistant May 2025 – Present
 - Work on MCTS for Orthopedic trauma triage system, pivot into more methodological paper, Find a gap in the literature of Jaw cancer (Osteonecrosis)

- * **Dr. Gholamali Aminian** *Remote*
Research Intern September 2025 – August 2026
 - Investigating feature extracted from battle data to predict Human Preferences

RESEARCH PROPOSALS

- **QuantaSR: Efficient Image Super-Resolution through Quantization Error Optimization in Low-Precision Neural Networks.** Farzan Niknejad, Mohammad Hadi Niknam, and Javad Alirezaie. ICML 2027.
- Efficient Super Resolution Using Multi-Task Learning with Conflict Aware optimization (MLTSR)
- TailRL extention : Environmental aware Weighting for Best-of-K weighting in TailRL
- VL-JEPA for EchoCardio
- BoneSAM, Finetuning MedSAM on BTXRD
- SonoGym+ : Extending SonoGym to become pressure aware environment
- VQA benchmark development for the BTXRD dataset

WORK EXPERIENCE

- **ResanAI** *Tehran*
AI Research Engineer March 2026 – June 2026
- Completed an AI engineering internship focused on graph algorithms and Graph Neural Networks (GCN, GraphSAGE, and EdgeConv, a graph-based architecture and geometric algorithms in point cloud literature, Study PyG library internals

SELECTED COURSES

- PDE : (18.5/20), Signal&System : (18/20), Deep RL 2025 (audited), MultiAgent RL 2026 (audited)

HONORS AWARDS

- Gold medalist in Iran national Economics Olympiad
- **Iran University Entrance Exam** : top 1% over 100,000 Iranian students compete for top universities.

LANGUAGES



Persian: Native Speaker



English: Professional Proficiency — Official TOEFL iBT: 90 / 120 (Verified Score in Hand)



German: Limited working Proficiency (B1) "*Goethe Institute Teheran*"



French: Limited understanding (A2)



Arabic: Basic understanding

REFERENCES

Dr. Alireza Esmailzahi

Postdoctoral Fellow, University of Toronto, Canada

Email: alireza.esmailzahi@utoronto.ca

Dr. Farzan Niknejad Mazandarani

Postdoctoral Fellow, Toronto Metropolitan University, Canada

Email: fniknejad@torontomu.ca

Dr. Mohammad Farid Azampour

Postdoctoral Researcher, Technical University of Munich, Germany

Email: mf.azampour@tum.de

Dr. Mohammad Reza Dolikhani

Medical Doctor (MD) – Researcher, Shahid Beheshti University of Medical Sciences, Iran

Email: mohammadrezadolikhani@sbmu.ac.ir

Dr. Javad Alirezaie

Professor, Toronto Metropolitan University, Canada

Email: javad@torontomu.ca